Introduction to Linked Lists

Q. What is a Linked List? Explain its representation in memory with the help of a suitable diagram and structure definition in C.

> **Definition to Remember**

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1. Structure of a Node

A standard linked list node has two components:

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Structure Definition in C:

```c

## 2. Representation in Memory

- The list is accessed via a special starting pointer called the **`head`** (or `start`).

- If th

**Diagram:**

```text

[Data|Next]  [Data|Next]  [Data|Next]  NULL

(Node

## 3. Advantages over Arrays

- **Dynamic Memory Allocation:** Size can grow and shrink at runtime; memory is allocated exactly as needed (no wastage).

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>  **Must-Write Points (for 10 marks)**

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> **Quick Recall**

> `Link
struct